4.5 Destressing the Grid

Comparison-only version: original PID comparison, best observed results, and their trade-offs. This document contains only Section 4.5 and a newly assembled Appendix Q. It does not replace the finalized full manuscript. Appendix Q combines the relevant historical and newer supplementary evidence for this standalone version; the original archive itself ended at Appendix P.

What changed from the archived result. The original `camera_ready.tex` reported PID at 6.74 MW and its frozen SAC at 0.21 MW on three peak dates. Those reported legacy results are preserved verbatim in Table Q.1. Later work changed physical accounting and replay protocols. The electricity-billing fixes alone do not explain PID's MW change: a dedicated reproduction recovers 6.7385 versus 5.6408 MW and exposes different input trajectories and cooling responses (Appendix Q.2). Changing the SAC checkpoint/training coverage and adding two-hour recovery were separate experimental choices, not bug fixes. Therefore 0.21 to 6.007 MW is *not* a 29-fold algorithm-only improvement.

PID and low-level controller comparison. Table 2 retains PID, Rule and Bang-bang and adds the strongest low-level follow-ups under the original *rule macro* and peak-case labels. The corrected retrained SAC has 6.007 versus PID's 5.641 MW facility relief, 658.4 versus 821.9 kW tracking RMSE, and 648.4 USD more 26-hour cash, but uses 31.682 versus 29.344 MWh battery throughput. Analytic control and BC track more accurately; later regularized SAC variants exchange a small amount of relief/cash for much lower battery use. There is no single winner on every objective.

Table 2: Updated counterpart of the original Table 2, with PID retained. All rows use the **rule-based macro**, corrected `physical_v1`, and the original three peak dates, seeds 100–104, $\beta = 0.5$ (15 cases). Four-checkpoint methods are averaged within each case first; the retrained SAC row is one previously selected checkpoint. ± is descriptive case SD, not a confidence interval. This is historical regression evidence, not an untouched test.

(a) Original grid-relief metrics

Controller	Facility relief	LFR	SPDR	LWSR	LW-CF	ΔR_β	Pass/planned
Rule	5.068 ± 1.482	5.03 ± 1.45	0.049 ± 0.014	0.020 ± 0.006	6.20 ± 1.92	−0.80 ± 1.05	**30/30**
PID†	5.641 ± 1.484	5.60 ± 1.46	0.054 ± 0.014	0.024 ± 0.006	7.45 ± 1.85	−1.60 ± 0.47	26/30
Bang-bang	4.701 ± 1.078	4.67 ± 1.06	0.045 ± 0.010	0.022 ± 0.006	6.65 ± 1.74	−3.03 ± 1.18	**30/30**
Retrained SAC	**6.007** ± 1.724	**5.97** ± 1.70	**0.058** ± 0.017	**0.030** ± 0.006	**9.09** ± 1.88	−0.34 ± 0.52	**30/30**
Analytic	5.626 ± 1.645	5.59 ± 1.63	0.054 ± 0.016	0.028 ± 0.007	8.54 ± 2.17	**−0.02** ± 0.03	**30/30**
BC only	5.583 ± 1.636	5.55 ± 1.62	0.054 ± 0.016	0.028 ± 0.007	8.52 ± 2.09	−0.26 ± 0.26	**120/120**
Guided SAC	5.607 ± 1.647	5.57 ± 1.63	0.054 ± 0.016	0.028 ± 0.007	8.64 ± 2.12	−0.33 ± 0.27	**120/120**
Union SAC + shaping	5.567 ± 1.632	5.53 ± 1.62	0.054 ± 0.016	0.028 ± 0.007	8.48 ± 2.09	−0.31 ± 0.26	**120/120**
Union SAC	5.582 ± 1.634	5.55 ± 1.62	0.054 ± 0.016	0.028 ± 0.007	8.51 ± 2.10	−0.28 ± 0.27	**120/120**

(b) Tracking, actual PCC relief, cash and physical checks

Controller	PCC relief	RMSE	BESS	Δcash/PID	Peak PCC24	Pass/planned
Rule	4.972 ± 1.635	784.2 ± 114.5	31.003 ± 2.216	−769.4 ± 219.9	1.225 ± 2.337	**30/30**
PID†	5.610 ± 1.537	821.9 ± 80.8	29.344 ± 1.495	0.0 ± 0.0	1.995 ± 2.153	26/30
Bang-bang	4.640 ± 1.178	1681.0 ± 255.9	40.171 ± 5.149	−871.9 ± 400.1	1.898 ± 2.215	**30/30**
Retrained SAC	**5.775** ± 1.678	658.4 ± 308.1	31.682 ± 2.024	**648.4** ± 140.7	**2.050** ± 2.184	**30/30**
Analytic	5.626 ± 1.645	**50.2** ± 74.4	0.950 ± 1.510	466.2 ± 173.8	2.005 ± 2.058	**30/30**
BC only	5.581 ± 1.636	290.5 ± 267.6	**0.109** ± 0.006	387.6 ± 165.9	2.003 ± 2.042	**120/120**
Guided SAC	5.612 ± 1.648	382.7 ± 239.3	0.719 ± 0.108	427.4 ± 179.4	1.988 ± 2.061	**120/120**
Union SAC + shaping	5.568 ± 1.632	295.3 ± 263.4	0.134 ± 0.007	364.5 ± 165.8	2.004 ± 2.038	**120/120**
Union SAC	5.579 ± 1.633	294.4 ± 264.1	0.172 ± 0.009	377.7 ± 167.9	2.008 ± 2.041	**120/120**

Relief and peak values: MW; RMSE: kW; BESS: absolute 26-hour throughput in MWh; cash: 26-hour USD versus matched PID under the same macro. LFR/SPDR/LWSR/LW-CF and revenue changes are percentages. ΔR_β compares each method to its own matched zero-bias run, not to PID. Pass counts include *both* biases (30 cases per checkpoint). †: a complete cohort containing physical/recovery gate failures, not an admitted cohort. No failed cases were dropped from the means. Bold marks best displayed qualified means at printed precision (lower RMSE/BESS; higher other metrics), not significance. SPDR is a stress-average proxy, not actual peak reduction.

How to read the progression. The corrected SAC–PID facility-relief gap on these matched cases is 0.366 MW. Separate panels below show whether this generalizes, what changes when the macro changes, and what the economic supervisor adds. Stress-window mean facility relief excludes battery exchange; PCC relief includes it. Actual peak reduction is a difference of maxima, not SPDR. Cash includes 24-hour service plus two-hour continuing recovery, not business profit. No table mixes biases or replaces a failed cohort by a survivor-only average.

Q Supplementary PID Comparisons and Result Progression

Q.1 The archived starting point

The following is the original Table 2 body from the ZIP's active `camera_ready.tex`, not a different `.tex` variant. Its original metric layout, means, standard deviations and emphasis are preserved. These are *reported legacy numbers*: no recovery qualification, newer cash metric or later model is retroactively attached to them.

Table Q.1: Original archived Table 2, reproduced as reported (not a new evaluation). Grid-relief results at $\beta = 0.5$: mean ± sample standard deviation over 15 evaluation episodes. Columns denote hardware controllers paired with a rule-based macro controller; "Rule" denotes rule-based control. ↑ indicates higher is better.

Metric↑	**Rule**	**PID**	**Bang-bang**	**SAC**
Grid relief				
Mean stress-window relief ($\overline{\Delta P}_S$, MW)	6.16 ± 1.33	$\mathbf{6.74 \pm 1.29}$	5.46 ± 0.78	0.21 ± 0.40
Load flexibility ratio (LFR, %)	7.35 ± 1.60	$\mathbf{8.04 \pm 1.55}$	6.52 ± 0.95	0.25 ± 0.48
System peak demand relief (SPDR, %)	0.059 ± 0.013	$\mathbf{0.065 \pm 0.013}$	0.053 ± 0.008	0.002 ± 0.004
Stress-weighted reduction (LWSR, %)	0.022 ± 0.011	$\mathbf{0.025 \pm 0.011}$	0.021 ± 0.009	-0.013 ± 0.004
Load-weighted capacity factor (LW-CF, %)	6.81 ± 3.30	$\mathbf{7.77 \pm 3.45}$	6.39 ± 2.72	-4.14 ± 1.20
Regulation revenue change (Δ Reg. Rev., %)	-3.13 ± 0.81	-4.17 ± 1.05	$\mathbf{-2.79 \pm 1.21}$	-22.53 ± 4.67
Facility				
Throughput reward change (Δr_{thr}, per tick)	-0.054 ± 0.012	-0.059 ± 0.013	-0.071 ± 0.012	$\mathbf{-0.018 \pm 0.001}$
SOC reward change (Δr_{soc}, per tick)	$\mathbf{-0.134 \pm 0.026}$	-0.151 ± 0.025	-0.174 ± 0.023	-0.146 ± 0.031
Thermal reward change (Δr_{therm}, per tick)	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	$\mathbf{0.000 \pm 0.000}$	-0.006 ± 0.010

The old SAC was not trained for the same biased-signal coverage used by later methods. The PID reference also moves from 6.74 to 5.641 MW under the corrected study, so version/protocol effects must not be hidden inside a claimed learning gain. The reward-component deltas in the archival table's facility block are not silently treated as physical work, cash or thermal-storage measurements.

Q.2 PID relief change under the revised protocol

The billing examples are not an explanation of the MW drop. Electricity-cost units affect cash accounting. Settlement duration for early-ended intervals does not account for these results: every reproduced episode completed all 17,280 service ticks and 96 macro intervals. The two-hour recovery occurs after the 24-hour service-relief measurement, and changing a SAC checkpoint cannot change a separately evaluated fixed PID.

Direct reproduction. We reran the three original date labels and five seed labels in each simulation mode with the same PID implementation and gains. There were 30 PID/nominal-reference pairs, hence 60 complete trajectories, with no training or retuning. The earlier mode reproduces 6.7385 ± 1.2854 MW; the corrected mode reproduces 5.6408 ± 1.4843 MW. These round to the two values in the comparison. The date/seed labels are held constant, but the replay and realized inputs are *not* held identical across modes.

Specific differences in the implementation and evaluation protocol.

1. **Calendar replay.** The old evaluator repeated each load record as though it represented five minutes. The three source days contain 302, 299 and 294 valid records, not 288 equal-duration records. Their first 288 records reach only 22:45, 23:00 and 23:30, respectively, despite a nominal 24-hour simulation. The newer replay uses actual timestamps. Weather is aligned by calendar fields rather than by shifting a row index; the 2024 weather remains calendar climatology, not contemporaneous 2023 observations.

2. **Workload phase.** The old date-selection path shifted load/weather but left the workload at phase 0. The newer path also advances the repeating 30-day workload to phases 27, 8 and 9 for the three dates. This changes operating conditions; it is not a PID algorithm improvement or added installed capacity.

3. **Random streams and causal market information.** RegD, offers, clearing and frequency use separate random streams in the newer replay. The same integer seed therefore no longer generates the same old/new regulation

and market trajectories. The current offer is exposed before the bid, and the newly cleared commitment before the hardware action.

4. **Physical bookkeeping.** Idle power is separated from deferrable work, and executed IT power, heat and electrical accounting are made consistent. These changes also affect the simulated plant and nominal-reference trajectory. The installed 250 MW plant is not enlarged.

These are a mixture of implementation corrections and protocol choices, not one isolated bug fix. In the actual reproduction, observed stress-window RegD falls from 0.5129 to 0.4110 and the requested response falls from 7.0430 to 5.6263 MW. PID consequently produces less cooling-related relief under the changed trajectories. The average HVAC command rises from 0.5697 to 0.5908 (less cooling-power curtailment).

Table Q.2: Observed inputs and operating powers in the reproduced PID runs. The same date and seed labels do not imply identical realized inputs.

Quantity (stress-window mean)	Earlier mode	Corrected mode	Change
Observed RegD (dimensionless)	0.5129	0.4110	-0.1018
Actual commitment (MW)	13.7793	13.3425	-0.4368
Requested response (MW)	7.0430	5.6263	-1.4167
Nominal-reference facility power (MW)	83.8546	100.6322	+16.7776
PID facility power (MW)	77.1161	94.9914	+17.8753
HVAC effort (0–1)	0.5697	0.5908	+0.0212

Where the reported 1.0977 MW difference appears. Cooling-related relief falls by 1.0984 MW; the remaining facility terms add 0.0007 MW. Their sum gives the 1.0977 MW decrease in total facility relief. The reference and controlled facility powers both change, as the table above shows; this is not a calculation in which only the electricity bill changed.

Table Q.3: Observed facility-relief power balance. Cooling and remaining facility terms sum to the total; these are not per-fix causal effects.

Quantity (stress-window mean)	Earlier mode	Corrected mode	Change
Cooling-related relief (MW)	6.7178	5.6194	-1.0984
Remaining facility relief (MW)	0.0207	0.0214	+0.0007
Total facility relief (MW)	6.7385	5.6408	-1.0977

Attribution limit. This is an exact observed power-component decomposition, *not* a one-change-at-a-time causal ablation. We have not assigned an independent MW share to timestamp correction, workload phase, random-stream changes, market timing or plant bookkeeping. Doing so would require holding the other components fixed. It would be incorrect to say that PID lost 1.10 MW because electricity billing was fixed, or that its algorithm became worse under otherwise identical inputs.

Q.3 Low-level tracking: what replicated and what did not

Final-B uses six different dates (13 January, 7 April, 18 July, 7 August, 21 September and 11 November 2023), three shared noise labels per date and both biases. These cases were reserved in that earlier study and are consumed now. The results below are *not* the eight-date economic-confirmation panel. On Final-B, BC's 53.3 kW RMSE is better than conditioned Union SAC's 58.3 kW, although both improve substantially on the earlier Guided checkpoint. The retrained SAC's peak-regression cash advantage does not replicate here: its mean cash change versus PID is negative. Extra training did not establish an incremental online-RL advantage over BC.

Table Q.4: Tracking-only Final-B, rule macro, $\beta = 0.5$, 18 date/noise cases. Means/SDs average four checkpoints per case where applicable. Physical/recovery pass counts include both biases (36 cases per checkpoint).

Controller	PCC relief	RMSE	BESS	Δcash/PID	Peak PCC24	Pass/planned
Rule†	5.323 ± 1.324	693.2 ± 121.2	32.780 ± 4.172	-300.4 ± 178.0	3.950 ± 2.816	34/36
PID†	5.822 ± 1.288	715.1 ± 103.9	31.099 ± 4.076	0.0 ± 0.0	4.305 ± 2.888	34/36
Bang-bang†	5.172 ± 1.036	1221.7 ± 256.7	45.729 ± 7.557	-289.0 ± 274.7	3.905 ± 3.468	28/36
Retrained SAC	5.671 ± 1.288	431.3 ± 134.7	42.987 ± 5.331	-32.1 ± 426.9	4.067 ± 2.978	**36/36**
Analytic	5.852 ± 1.139	$\mathbf{0.2} \pm 0.1$	$\mathbf{0.002} \pm 0.008$	$\mathbf{268.6} \pm 119.2$	4.417 ± 2.846	**36/36**
BC only	5.836 ± 1.135	53.3 ± 12.1	0.092 ± 0.005	243.4 ± 113.4	4.421 ± 2.842	**144/144**
Guided SAC	$\mathbf{5.860} \pm 1.142$	181.4 ± 17.3	0.834 ± 0.142	250.8 ± 123.6	$\mathbf{4.430} \pm 2.841$	**144/144**
Union SAC + shaping	5.830 ± 1.134	58.3 ± 12.0	0.142 ± 0.011	235.6 ± 110.7	4.419 ± 2.842	**144/144**
Union SAC	5.829 ± 1.136	59.1 ± 11.6	0.129 ± 0.009	234.5 ± 111.2	4.417 ± 2.841	**144/144**

Units: PCC/peak MW; RMSE kW; BESS MWh; cash USD versus matched PID. †: complete but not fully qualified cohort; no survivor filtering. All low-level reward and gate decisions remain those of their original study.

Q.4 High-level changes: separate the macro, executor and learned allocation

First hold hardware fixed and change only the existing market rule. This is an observed macro benefit, not a learned-bidding result or a proof that maximum quantity is universally optimal.

Table Q.5: Changing the macro alone on the same original peak cases. Hardware identity and evaluation cases are matched within each row. The full-residual zero-ask macro is an existing rule, not learned bidding.

Hardware	Rule relief MW	Full-res. relief MW	Δcash USD	Rule RMSE kW	Full-res. RMSE kW
PID	5.641	7.581	3796.1	821.9	755.7
Retrained SAC	6.007	8.127	3904.8	658.4	686.3
BC only	5.583	7.790	3967.9	290.5	363.6
Guided SAC	5.607	7.773	3866.0	382.7	460.0
Union SAC + shaping	5.567	7.749	3951.2	295.3	348.1

These are complete-case descriptive means, not superiority tests or a proof that maximum quantity is globally optimal. Gate counts remain visible in the source panels.

Next keep the full-residual zero-ask macro fixed on the *same original peak dates and seeds 100–104*. The following end-to-end comparison retains the matched historical PID reference and shows the economic stages. The low-level rows directly actuate devices; the economic rows share an analytic executor and allocate additional response. Gains between these architectures combine executor and allocation changes, rather than isolating the SAC update. The earlier absolute-MW rules and newer fraction rules are labelled separately.

Table Q.6: Original peak regression under the full-residual macro, $\beta = 0.5$ (15 matched date/noise cases). No new-noise confirmation averages are substituted into this panel. Cash is relative to the PID row.

Controller	Facility	PCC	RMSE	BESS	Δcash/PID	Pass/planned
PID	7.581	7.552	755.7	32.064	0.0	**30/30**
Retrained SAC	8.127	7.794	686.3	31.149	757.2	**30/30**
BC only	7.790	7.789	363.6	0.109	559.4	**120/120**
Guided SAC	7.773	7.789	460.0	0.760	497.4	**120/120**
Union SAC + shaping	7.749	7.753	348.1	0.180	519.6	**120/120**
Earlier economic PPO	8.694	8.694	1236.0	0.020	1686.9	**120/120**
Earlier economic SAC	8.782	8.782	1351.1	**0.000**	1793.9	**120/120**
Earlier absolute greedy	8.782	8.782	1351.8	**0.000**	1794.7	**30/30**
Earlier absolute margin 10	8.782	8.782	1379.1	**0.000**	1799.9	**30/30**
Analytic, no extra	7.351	7.351	**0.2**	0.152	357.1	**30/30**
Full-fraction rule	8.782	8.782	1351.8	**0.000**	1794.7	**30/30**
Deadline-margin rule	8.782	8.782	1371.3	**0.000**	1798.5	**30/30**
Credit-threshold rule	**8.878**	**8.878**	2736.6	0.105	1779.3	**30/30**
$f = 0.50$, margin 20	8.781	8.781	1466.8	0.034	1844.1	**30/30**
Fractional PPO	8.694	8.694	1169.9	0.040	1626.7	**120/120**
Fractional SAC	8.770	8.770	1319.1	0.001	1738.0	**120/120**
PPO + credit Φ	8.402	8.402	881.9	0.067	1312.5	**120/120**
SAC + credit Φ	8.825	8.825	1457.5	0.020	**1868.2**	**120/120**

Means only; no paired uncertainty is inferred from marginal SDs. Units: relief MW, RMSE kW, BESS MWh, cash USD. Gate counts include both biases; historical tracking rows use their recorded physical/recovery gates, while economic rows also use their recorded service floor. These are source-study gate counts, not a new uniform admission test.

Q.5 Newest results with newly matched PID references

The sixteen frozen neural checkpoints and 27 scripts were previously evaluated on eight new calendar combinations and new stochastic streams on three known peak dates. No weights have changed since that confirmation. For this extract, we subsequently ran **440 additional PID episodes**: the native controller and a separately labelled variant that only forces the executed batch throttle to one, each on exactly the same 220 cells. Gains and the other actuator commands are unchanged. No credit-allocation guard is retrofitted to PID. These PID additions are **post-confirmation descriptive references**, not new blinded evidence or an alteration of the original 43-policy primary test. The economic checkpoints in the original-peak panel above and this newer confirmation are also the same frozen weights: changed absolute averages under new noise streams are not additional training improvements.

All 440 episodes completed and recovered. Each PID variant passed 177/220 cases. The same 43 cells per variant miss only the *24-hour service-end SOC reserve* of 20%. Every case meets the 90% paid-service floor and closes recovery, with recharge electricity included in cash. This is *not* unreturned battery energy or unpriced recovery. Every printed PID mean still includes **all 80 or all 30 cases** in its stratum, including reserve failures; 56/80 is a qualification flag, not the averaging denominator. No admissible full-cohort PID mean exists under the strict all-cases gate, so PID rows are explicitly diagnostic (†).

Compare PID (full work) with Analytic, no extra to see the executor difference; compare Analytic, no extra with the fraction supervisors to see the added allocation layer. The full pipeline gap is not all attributable to learning. Native PID additionally has throttle freedom, and its actual work effects are measured separately below.

Table Q.7: Eight-date calendar panel, full-residual macro, $\beta = 0.5$; 80 matched complete cases per policy. Neural rows are four-checkpoint means. New PID references are descriptive additions.

Controller	Facility	PCC	RMSE	BESS	Δcash/PID	Service	Peak PCC26	Pass/planned
PID (native)†	7.031	7.174	788.7	31.794	0.0	97.67	4.807	56/80
PID (full work)†	7.033	7.176	733.5	31.794	2.1	97.69	4.807	56/80
Analytic, no extra	7.083	7.083	**0.2**	**0.000**	302.9	**100.00**	4.737	**80/80**
Full-fraction rule	8.307	8.307	1189.2	**0.000**	494.1	92.01	5.964	**80/80**
Deadline-margin rule	8.390	8.390	2101.9	**0.000**	**619.4**	92.05	6.068	**80/80**
Credit-threshold rule	8.273	8.273	2595.6	**0.000**	498.6	92.16	4.963	**80/80**
Fractional PPO	8.196	8.196	1005.2	**0.000**	476.2	93.45	5.852	**320/320**
Fractional SAC	8.237	8.237	1160.3	**0.000**	487.1	92.63	5.882	**320/320**
PPO + credit Φ	7.977	7.977	769.8	**0.000**	435.7	95.10	5.634	**320/320**
SAC + credit Φ	**9.172**	**9.172**	1612.4	**0.000**	537.5	92.02	**6.887**	**320/320**

Relief/peak: MW; RMSE: kW; BESS: 26-hour MWh; Δcash/PID: complete-case 26-hour USD versus native PID; Service: mean actual paid-service percentage. †: complete but fails the all-cases SOC-reserve admission rule; the cash difference is descriptive, not a new confirmed superiority claim over an admitted PID cohort. Pass counts apply to the displayed bias. Bold excludes diagnostic cohorts and is not significance. Zero-bias negative peak values must not be described as peak shaving.

Table Q.8: Eight-date calendar panel, full-residual macro, $\beta = 0.0$; 80 matched complete cases per policy. Neural rows are four-checkpoint means. New PID references are descriptive additions.

Controller	Facility	PCC	RMSE	BESS	Δcash/PID	Service	Peak PCC26	Pass/planned
PID (native)†	0.012	−0.006	607.7	56.193	0.0	98.65	−2.778	68/80
PID (full work)†	0.012	−0.006	600.0	56.193	0.2	98.65	−2.778	68/80
Analytic, no extra	0.011	0.011	**0.1**	**0.000**	106.3	**100.00**	−2.919	**80/80**
Full-fraction rule	1.234	1.234	1189.2	**0.000**	297.5	92.01	−1.682	**80/80**
Deadline-margin rule	1.318	1.318	2101.9	**0.000**	**422.9**	92.05	−1.601	**80/80**
Credit-threshold rule	1.201	1.201	2595.6	**0.000**	302.0	92.16	−2.702	**80/80**
Fractional PPO	1.138	1.138	1016.1	**0.000**	281.4	93.30	−1.797	**320/320**
Fractional SAC	1.180	1.180	1166.9	**0.000**	291.7	92.60	−1.744	**320/320**
PPO + credit Φ	0.925	0.925	777.6	**0.000**	240.9	95.01	−2.006	**320/320**
SAC + credit Φ	**2.275**	**2.275**	1626.7	**0.000**	345.3	92.02	**−0.554**	**320/320**

Relief/peak: MW; RMSE: kW; BESS: 26-hour MWh; Δcash/PID: complete-case 26-hour USD versus native PID; Service: mean actual paid-service percentage. †: complete but fails the all-cases SOC-reserve admission rule; the cash difference is descriptive, not a new confirmed superiority claim over an admitted PID cohort. Pass counts apply to the displayed bias. Bold excludes diagnostic cohorts and is not significance. Zero-bias negative peak values must not be described as peak shaving.

Table Q.9: Three known peaks, new noise, full-residual macro, $\beta = 0.5$; 30 matched complete cases per policy. Neural rows are four-checkpoint means. New PID references are descriptive additions.

Controller	Facility	PCC	RMSE	BESS	Δcash/PID	Service	Peak PCC26	Pass/planned
PID (native)†	8.692	8.695	756.2	31.009	0.0	97.81	5.850	27/30
PID (full work)†	8.692	8.695	756.2	31.009	0.0	97.81	5.850	27/30
Analytic, no extra	8.579	8.579	**0.2**	0.208	526.5	**100.00**	5.822	**30/30**
Full-fraction rule	10.010	10.010	1355.1	**0.000**	1966.6	92.00	7.255	**30/30**
Deadline-margin rule	10.010	10.010	1376.4	0.001	1970.7	92.00	7.255	**30/30**
Credit-threshold rule	**10.117**	**10.117**	2747.3	0.148	1966.0	92.08	**7.493**	**30/30**
$f = 0.50$, margin 20	10.009	10.009	1473.6	0.086	2016.0	92.09	7.254	**30/30**
Fractional PPO	9.917	9.917	1170.7	0.050	1792.2	93.33	7.162	**120/120**
Fractional SAC	10.001	10.001	1321.9	**0.000**	1909.4	92.32	7.243	**120/120**
PPO + credit Φ	9.626	9.626	881.9	0.083	1479.1	95.09	6.873	**120/120**
SAC + credit Φ	10.064	10.064	1462.2	0.017	**2038.1**	92.01	7.308	**120/120**

Relief/peak: MW; RMSE: kW; BESS: 26-hour MWh; Δcash/PID: complete-case 26-hour USD versus native PID; Service: mean actual paid-service percentage. †: complete but fails the all-cases SOC-reserve admission rule; the cash difference is descriptive, not a new confirmed superiority claim over an admitted PID cohort. Pass counts apply to the displayed bias. Bold excludes diagnostic cohorts and is not significance. Zero-bias negative peak values must not be described as peak shaving.

Table Q.10: Three known peaks, new noise, full-residual macro, $\beta = 0.0$; 30 matched complete cases per policy. Neural rows are four-checkpoint means. New PID references are descriptive additions.

Controller	Facility	PCC	RMSE	BESS	Δcash/PID	Service	Peak PCC26	Pass/planned
PID (native)†	0.033	−0.110	683.4	60.399	0.0	98.56	−3.110	26/30
PID (full work)†	0.033	−0.110	683.4	60.399	0.0	98.56	−3.110	26/30
Analytic, no extra	−0.025	−0.025	**0.1**	**0.000**	405.1	**100.00**	−3.137	**30/30**
Full-fraction rule	1.406	1.406	1355.1	**0.000**	1845.3	92.00	−1.703	**30/30**
Deadline-margin rule	1.406	1.406	1376.4	**0.000**	1849.4	92.00	−1.703	**30/30**
Credit-threshold rule	1.513	1.513	2747.3	**0.000**	1844.7	92.08	**−1.465**	**30/30**
$f = 0.50$, margin 20	1.405	1.405	1473.6	**0.000**	1894.7	92.09	−1.705	**30/30**
Fractional PPO	1.333	1.333	1186.2	**0.000**	1689.1	93.16	−1.793	**120/120**
Fractional SAC	1.390	1.390	1316.2	**0.000**	1778.3	92.37	−1.725	**120/120**
PPO + credit Φ	1.038	1.038	888.4	**0.000**	1368.1	95.02	−2.077	**120/120**
SAC + credit Φ	**1.514**	**1.514**	1562.0	**0.000**	**1996.0**	92.01	−1.615	**120/120**

Relief/peak: MW; RMSE: kW; BESS: 26-hour MWh; Δcash/PID: complete-case 26-hour USD versus native PID; Service: mean actual paid-service percentage. †: complete but fails the all-cases SOC-reserve admission rule; the cash difference is descriptive, not a new confirmed superiority claim over an admitted PID cohort. Pass counts apply to the displayed bias. Bold excludes diagnostic cohorts and is not significance. Zero-bias negative peak values must not be described as peak shaving.

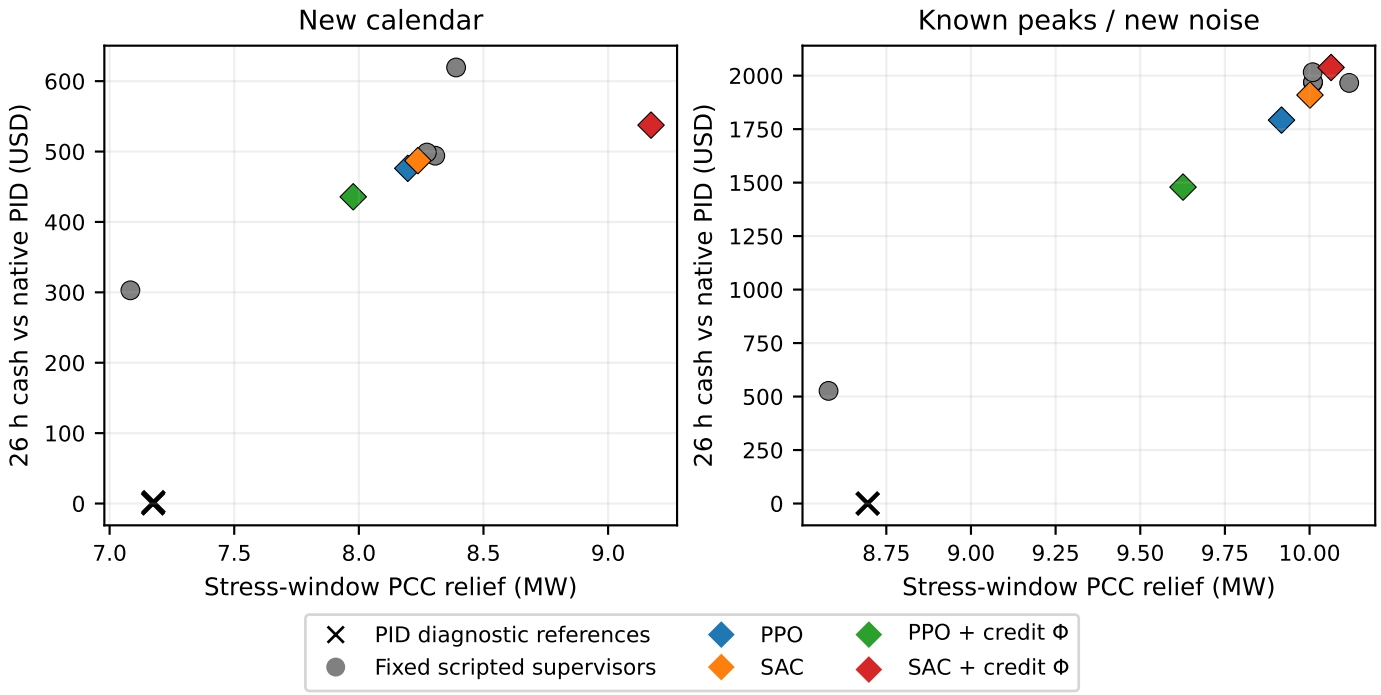


Figure Q.1: Matched $\beta = 0.5$ progression in relief and 26-hour cash. PID crosses are full-cohort diagnostic references, not admitted policies. Diamonds are four-checkpoint means, not a deployed action-averaging ensemble. Selected scripts include the fixed full-fraction comparator and observed best-cash/relief examples.

Table Q.11: PID resource and reserve diagnostics, without survivor filtering. Positive IT curtailment is an electrical-energy difference, not a claim that the same amount of work energy was deferred.

Panel/β	PID variant	Stress throttle fraction	Stress IT curtail. MW	Positive IT curtail. MWh	Worst queue kWh	Min. end-SOC %	Pass /planned
New calendar/0.0	Native	0.9928	0.000	0.004	258.12	12.02	68/80
New calendar/0.0	Full work	1.0000	0.000	0.000	0.00	12.02	68/80
New calendar/0.5	Native	0.8369	-0.002	0.042	983.97	12.00	56/80
New calendar/0.5	Full work	1.0000	0.000	0.000	0.00	12.00	56/80
Known peak/0.0	Native	0.9889	0.000	0.000	0.00	12.02	26/30
Known peak/0.0	Full work	1.0000	0.000	0.000	0.00	12.02	26/30
Known peak/0.5	Native	0.7959	0.000	0.000	0.00	11.99	27/30
Known peak/0.5	Full work	1.0000	0.000	0.000	0.00	11.99	27/30

All 440 episodes complete service and recovery and meet the 90% paid-service floor. The only failures are the 24-hour service-end SOC reserve of 20%; all recover to initial SOC, and recovery charging is priced in the reported cash. Throttle below one does not by itself establish deferred work; the queue and actual IT-meter columns are reported separately.

The full-work override leaves the peak-panel relief, tracking and cash identical in all 60 matched cases. Calendar-panel means differ only slightly; both variants fail the identical 43 service-SOC reserve checks. The extra throttle freedom is therefore not interchangeable with actual work deferral or a demonstrated aggregate economic benefit.

Q.6 What the comparison does and does not establish

- **Low-level tracking improved, but not every improvement is RL.** The original matched regression now has a corrected retrained SAC ahead of PID on facility relief, RMSE and cash, with more battery use. BC and analytic control remain better trackers; Union SAC's extra training does not beat BC on Final-B. Warm SAC was not admitted; the single-source state-local pilots did not earn their extension.
- **The economic result is a different quality trade-off.** SAC with credit potential passed its original comparison against the fixed full-fraction rule: +0.866 MW PCC relief and +43.44 USD on the biased eight-date panel, with conditional intervals $[0.836, 0.896]$ MW and $[41.21, 45.66]$ USD. These condition on the fixed dates and four

checkpoints. Per-checkpoint relief gains range from 0.264 to 2.367 MW; the mean is not a typical-seed guarantee. The best primary-panel cash script still earns 81.9 USD more with 0.782 MW less relief. On known peak dates, the corresponding full-fraction contrasts are only 0.053 MW and 71.54 USD.

- **PPO was tested, not omitted.** Native low-level PPO and SAC each trained for 552,960 physical transitions per seed; their development qualification counts were 0/96 and 88/96, and neither passed the required all-seed BC comparison. Economic PPO/SAC and their fraction/potential variants each used 24,576 macro transitions per checkpoint. The potential-SAC family is the stronger economic candidate here; that does not turn the other families' results into successes.
- **Do not mix the references.** Relief is against independent nominal operation. Historical revenue change compares each controller to its own $\beta = 0$ run. This extract's cash columns compare to PID within the same panel/macro; the prior confirmation table used analytic no-extra as its cash reference. Monetary references are not relief baselines, and cash is not business profit.
- **Do not equate different protocols.** The archived legacy table, corrected rule-macro comparison, full-residual economic stages, Final-B dates and new-noise confirmation are separate panels. NYC 2023 load is paired with month/day-aligned 2024 weather under the frozen loader. Workload phases repeat. The original unused six-date reservation remains unopened. New PID numbers do not rewrite any previous gate or prespecified inference.

Q.7 Source and audit trail

Only the active `camera_ready.tex` member of the original ZIP supplied the archived table. Canonical corrected, Union, economic and confirmation reports supplied the later panels. New PID outputs are independently checked for metering, work conservation, paid-service ratio, state closure and exact exogenous/reference pairing. The original confirmation remains 43 policies and 9,460 cases, unchanged. Five independent adversarial reviewers examined the PID reference design, provenance, physical accounting, statistics and presentation. Machine-readable comparison values and source hashes accompany this PDF in the local study directory. The separate PID mode reproduction adds 60 diagnostic trajectories, not new confirmation cases; its summary, complete trace inventory and exact probe source are preserved. No policy weights or gains were changed.